Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	syntax
Project / Solution Name	CareerLens
Name (Leader)	'Ainaa Qistina Binti Mohd Nazrin
Email & Phone (Leader)	ainaaqistinanzrin@gmail.com | 01116220071
University Affiliation(s)	Universiti Teknikal Malaysia Melaka (UTeM)
Live Prototype URL	https://CareerLens.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/ainaaqstina/SDC_Hackathon_Submission_syntax

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

CareerLens adopts a modular web-based architecture integrating AI powered skills analysis, evidence-based career matching, and personalised career development.

The frontend uses React, TypeScript, Vite, and Tailwind CSS, while the Node.js and Express backend manages user profiles, CV processing, career recommendations, and learning pathways.

Uploaded CVs are processed through text extraction and a configured AI model hosted on RunPod. Users can review their extracted information before saving confirmed profiles in PostgreSQL through Drizzle ORM.

A Python-based matching engine compares user capabilities against structured occupational requirements, including O*NET occupation and skills data. It calculates explainable career fit scores using software relevance, transferable skill coverage, and personal interests.

CareerLens also integrates the TheirStack job market data connector through Replit to retrieve job listings and employer skill requirements, supporting evidence -based career recommendations and labour market insights.

The platform supports importing occupational datasets, retrieving job market information through an authorised Adzuna integration, and maintaining dated market observations. CareerLens also provides skills gap explanations, reskilling recommendations, future scenario comparisons, and SDG associations.

Replit serves as the primary development and deployment environment, while RunPod provides AI inference infrastructure. External datasets and market sources support transparent, evidence based recommendations.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9	Frontend and backend application logic	Apache-2.0
Node.js 24	Backend JavaScript runtime	MIT
React 19.1.0	Interactive user interface	MIT
Vite 7.3.7	Frontend development and build	MIT
Tailwind CSS 4.3.3	Responsive styling	MIT
Express 5.3.x	Backend REST APIs	MIT
PostgreSQL 16.10	Relational data storage	PostgreSQL License
Drizzle ORM 0.45.4	Database management	Apache-2.0
Zod 3.25.76	API input validation	MIT
Python 3.11.14	Supporting career-matching component	PSF License
pdf-parse 2.4.x	Extracting text from PDF CVs	MIT
Mammoth 1.13.x	Extracting text from DOCX CVs	BSD-2-Clause
Radix UI components	Accessible interface components	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod AI Inference API	Runs the configured AI model for CV analysis, career guidance and personalised learning recommendations	Sponsor-provided credits
Adzuna Job Search API	Retrieves occupation categories and job listing information for market data updates	Free
TheirStack Job Search API	Retrieves job postings, employer requirements, and hiring signals to support career matching, skills gap analysis, and job-market insights.	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Core	Web application development, testing, and intended deployment	Official SDC provided Replit Core subscription / Region not specified
RunPod	AI inference and model execution	USD 25 SDC sponsored credits / Region not specified
PostgreSQL hosting provider	Stores skills profiles, career data, roadmaps and trend observations	Replit managed PostgreSQL / Region not specified
GitHub	Source code hosting, version control and access for judges	GitHub Free / Cloud hosted repository

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3-8B	Qwen / Alibaba Cloud, served through RunPod	Pre-trained model for CV information extraction, career recommendation assistance and personalised learning pathways	Apache License 2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name				Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Dataset Name	Source / URL	Type	License	Personal Data?			
O*NET 31.0 Database	O*NET Resource Center	Public	CC BY 4.0	No			

Dataset Name				Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Future of Jobs Report 2025	World Economic Forum	Public research reference	WEF publication terms	No			
CareerLens Synthetic Profiles	Team created demonstration data	Synthetic	Team owned	No			
Job Listings: Open Jobs in Every Field (Daily)	https://www.kaggle.com/datasets/chandranshbinjola/job-listings	Public	Not verified (license information pending confirmation from Kaggle dataset publisher).	No			
Future Jobs & Skills Demand 2025	https://www.kaggle.com/datasets/ahsanneural/future-jobs-and-skills-demand-2025	Public	CC0: Public Domain	No			

Dataset Name				Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Global AI Jobs 2026 Dataset	https://www.kaggle.com/datasets/datascikhan/global-ai-jobs-2026	Public	CC0: Public Domain	No			

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT (OpenAI)	Project ideation, feature planning, technical guidance, debugging assistance, documentation and proposal writing	Major
Replit AI	Assisting with web application development, code generation, debugging and component integration	Major, if used extensively
Manus AI	Assisting with web application development, code generation, debugging and component integration	Minor, for trial web design

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Generative AI tools were used to support project planning, software development, debugging, and documentation. The team remains responsible for reviewing and validating all AI assisted outputs. Generated code is assessed through application testing, debugging, and verification against project requirements. Career recommendations are designed to rely on identifiable occupational data and transparent matching criteria rather than unrestricted AI generated claims. The team also plans to validate learning resources, dataset references, and SDG mappings before final deployment. AI generated suggestions are treated as supporting recommendations, while final design, implementation, and submission decisions remain the responsibility of the team members.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core subscription	Developing, testing, debugging, and preparing the CareerLens web application for deployment
RunPod	US$25 cloud credit	Providing AI inference infrastructure for Qwen3-8B to support CV analysis and personalised career guidance

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit and RunPod play important roles in the development and operation of CareerLens. Replit provides an integrated environment for application development, testing, and deployment, while RunPod supports AI model inference for skills extraction and personalised recommendations. RunPod is important to the current AI-assisted workflow, although the underlying model could potentially be hosted on alternative compatible infrastructure. Similarly, Replit could be replaced by another cloud development or hosting platform with appropriate configuration changes. These sponsor resources therefore improve accessibility and implementation feasibility without making the system permanently dependent on a single provider.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

CareerLens processes CV content, qualifications, skills, interests, and career preferences to provide personalised recommendations.

Uploaded CVs are converted into text and processed through the configured RunPod AI service. Original documents and extracted raw text are not intentionally stored after analysis. Users review and confirm structured profiles before information is saved in PostgreSQL.

Saved profiles and learning roadmaps can be removed through the application's deletion functionality.

API credentials are managed through server side environment variables, while administrative data-import operations require a configured access token. Synthetic profiles are used for hackathon demonstrations to avoid exposing real personal information.

The prototype does not yet establish a fixed automatic data-retention period. Additional authentication, access-control, privacy, and security testing will be required before public production deployment.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

CareerLens faces limitations involving AI reliability, occupational data coverage, and external service availability.

The AI model may incorrectly interpret CV information or produce inaccurate explanations. To mitigate this, extracted profiles are user reviewable, while career fit scores are computed using structured occupational evidence.

O*NET provides occupational information but does not necessarily represent current Malaysian hiring requirements. Adzuna job listing data may also be limited by geographical coverage, API availability, and provider restrictions.

Some occupations may not yet have complete SDG associations, and AI disruption scenarios remain illustrative estimates rather than guaranteed forecasts.

The system includes automated tests and source attribution mechanisms, but recommendation accuracy and fairness require further validation.

CareerLens also depends on external AI infrastructure and database availability. Manual profile entry and synthetic demonstration profiles provide alternatives when selected services are unavailable.

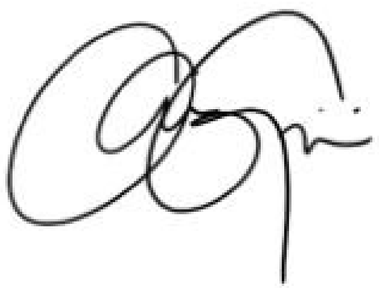
7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



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Date: 25/10/2026